

Inspect Five Parts with Calipers SOP

One episode inspects five parts. It runs from five parts standing in a numbered rack, an empty caliper anvil, two empty bins and a blank log sheet, to all five parts measured two ways each, fifteen marks on the log sheet, three parts in the pass bin, two parts in the fail bin and every tool back on its spot.

One part is finished before the next one is started. The order for each part never changes: **stand the part on the anvil, measure the tall way, measure the wide way, mark the log sheet, drop the part in its bin.** No part is measured until the caliper has been checked at zero, and no part goes in a bin until both of its readings are marked.

One arm does all of it. The same gripper takes every part out of the rack, stands it on the anvil, works the caliper slider, marks the log sheet and drops the part in its bin. Nothing in this task has to be held, because nothing moves on its own: the rack, the caliper stand, both bins and the log sheet all grip the table, so one gripper can pick out of them and drop into them with no second hand. Only one arm works in this cell. Do not bring a second arm into the task.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one part is held at a time, no grasp is changed while a part is off the table, and no part is ever carried over a bin it is not going into. A pass part leaves the anvil to the right. A fail part leaves the anvil to the left.

Each tool goes back to its own spot before the next step starts. The counts are fixed: five parts, one zero check, two readings per part, fifteen marks, three parts in the pass bin and two parts in the fail bin. There is no second zero check and no second reading of the same dimension. A part that lands off the table is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 90 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the parts rack, the caliper stand, the anvil, the good size card, both bins, the log sheet and the marker holder.

3. The caliper display faces the environment camera and stays readable in frame the whole time, and all five rows of the log sheet stay in frame the whole time.
4. The anvil is visible from above and from the side, so a reviewer can see the part standing on it and the jaw coming down on the part.
5. The arm is at home with its gripper open.
6. The tabletop is dry and clear of anything but the items in the Materials checklist.
7. The arm reaches every spot on the table, all five rack slots, the anvil, the slider knob at its top and at its bottom, both bins, the log sheet and the marker holder, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
8. The arm reaches every one of the five rack slots straight on, so a part can be lifted straight up out of any slot without leaning it over.
9. The paths from each rack slot to the anvil, from the anvil to each bin, and from the marker holder to every row of the log sheet are unobstructed. Carry one part along each path by hand during setup: it must clear the rim of every bin it passes and it must not touch the caliper.

Materials checklist

1. One parts rack stands at the back of the table, straddling the midline, with five upright slots in one row, numbered 1 on the left through 5 on the right. The numbers face the camera.
2. The rack grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or replace it. Nothing pins the rack during the episode.
3. Each slot holds one part standing upright, tall way up, with enough clearance that the part lifts straight out without rubbing.
4. Five parts, and no more. Each part is a plain rectangular block, clearly taller than it is wide, with flat square ends, no burrs and no markings on it.
5. The five parts are the same five in every episode: three good parts, one part that is over size the tall way, and one part that is under size the wide way.
6. The two bad parts are bad by a plain amount, far enough out that the reading lands well inside an end box on the strip and not on a box line. A part that is only just out is not used.
7. One caliper stand sits at the middle of the table, straddling the midline. It is heavy and it grips the table.
8. One digital caliper is clamped upright in the stand, beam vertical, with its fixed jaw resting level on the anvil and its moving jaw above, facing down. The clamp is tight: the caliper does not turn or slip in the stand when the slider is worked.
9. One anvil, a flat level plate at the foot of the caliper, big enough for a part to stand upright on it and to lie on its side on it with room to spare. It is bare and dry.
10. One slider knob is fitted to the caliper slider, big enough for the gripper to close on square, and it moves the slider up and down smoothly with light effort.

11. The caliper is switched on, set to the same units in every episode, and zeroed by hand during setup with its jaws shut. Nothing on this station presses a caliper button during the episode.
12. The slider starts the episode raised, with the moving jaw parked high enough above the anvil that the tallest part passes under it without touching.
13. One good size card lies flat on the table beside the caliper stand, printed with the good range for the tall way and the good range for the wide way. It is never picked up and never moved.
14. Two bins stand in their holders on the table: the pass bin right of the caliper stand and the fail bin left of the caliper stand. They are empty and dry, they are clearly different in colour, each is labelled PASS or FAIL facing the camera, and the same bin is the pass bin in every episode.
15. Each bin is low enough and wide enough that a part can be dropped in from just above the rim, and each one sits in its holder so it cannot tip or slide when a part lands in it.
16. One log sheet lies flat at the front centre on the midline. It is a wipe clean sheet with five rows, numbered 1 to 5 down the left, matching the five rack slots, and it starts the episode blank.
17. Each row of the log sheet holds a TALL strip of five boxes, then a WIDE strip of five boxes, then two result boxes labelled PASS and FAIL. Every box on a strip has a size range printed in it, the ranges run small on the left to large on the right, and they do not overlap.
18. On every strip the middle three boxes are printed with a heavy outline and the two end boxes are printed plain. The heavy boxes are the good sizes and they match the good size card.
19. Every box is big enough to hold one X with room to spare, and the boxes do not touch each other.
20. The log sheet grips the table. Push it by hand during setup: if it slides, tape it down or put it in a holder.
21. One dry erase marker stands in its holder on the log sheet's right edge, cap off, with enough ink to draw fifteen marks.

Workspace layout

Three rows: the parts at the back, the caliper stand in the middle with a bin on each side of it, and the log sheet at the front. Every part travels forward out of the rack to the anvil, then sideways off the anvil into a bin.

Back row:

1. Parts rack, back centre on the midline, five upright slots in one row, slot 1 on the left through slot 5 on the right, with the numbers facing the camera.

Middle row, left to right:

2. Fail bin spot, left of the caliper stand, the bin standing in its holder. A fail part leaves the anvil to the left.
3. Caliper stand, centre, straddling the midline, with the anvil at the foot of the caliper and the display facing the camera. The anvil is bare at the start of the episode and bare again at the end.
4. Pass bin spot, right of the caliper stand, the bin standing in its holder. A pass part leaves the anvil to the right.
5. Good size card, flat on the table between the caliper stand and the pass bin.

Front row:

6. Log sheet, front centre on the midline, its five rows facing the camera.
7. Marker holder, on the log sheet's right edge.

A clear gap of at least a gripper's width sits between the anvil and each bin, and between the caliper stand and the good size card, so a part drops into a bin straight from above without the gripper touching the caliper or the card. No carry passes over the bin the part is not going into.

Vocabulary

- **Arm assignment:** one gripper does everything: every part, the slider, every mark. This is a single arm task and no second arm takes any part in it.
- **Tool home:** the spot a tool returns to and stands in before the next step starts: the marker holder. A tool is never left on the log sheet, on the anvil, in a bin or in the gripper between steps.
- **Release point:** the gripper opens on the anvil when a part is being set down, just above the rim of a bin when a part is being dropped in, on the slider knob when the reading is being taken, or over a tool home. Never over the wrong bin and never over open tabletop.
- **Stable placement:** the object stands still after release, flat on its surface, and does not rock, tip, slide or lean on anything.
- **Part:** one plain rectangular block, taller than it is wide. There are five in every episode, one to a rack slot.
- **Tall way:** the part standing upright on the anvil, on one of its small flat ends, so the jaw comes down on its long side. This is the first of the two dimensions.
- **Wide way:** the part lying on its side on the anvil, so the jaw comes down across its short side. This is the second of the two dimensions.
- **Anvil:** the flat plate at the foot of the caliper that the part stands or lies on while it is measured. Only one part is ever on it.
- **Slider knob:** the knob on the caliper slider. It is the only part of the caliper the gripper ever touches.

- **Jaw down:** the slider lowered until the moving jaw meets the top of the part and stops. The jaw rests on the part. It is never pushed past that.
- **Jaw parked:** the slider raised so the moving jaw stands clear above the tallest part, with nothing between the jaws and the anvil.
- **Settled reading:** the number on the caliper display has stopped changing and holds still for 2 seconds with nothing touching the caliper, the stand or the part.
- **Zero check:** the one time in the episode the jaws are closed on nothing and the display is read. It happens once, before the first part, and the display has to read zero.
- **Good box:** a box on a strip printed with a heavy outline. The middle three boxes on every strip are good boxes.
- **Plain box:** a box on a strip printed without the heavy outline, at either end.
- **Matching box:** the one box on a strip whose printed range holds the number on the display. One reading matches one box and never two.
- **Good size card:** the card beside the caliper stand printed with the two good ranges. It is looked at, never touched.
- **Row:** one line of the log sheet. Row 1 is for the part from slot 1, row 2 for slot 2, and so on to row 5.
- **Marked box:** a box on the log sheet holding its X, fully inside its lines and not touching the next box.
- **Pass part:** a part whose two X marks both landed in good boxes. It goes in the pass bin.
- **Fail part:** a part with at least one X in a plain box. It goes in the fail bin.
- **One part at a time:** the gripper takes one part out of the rack, measures it, marks it and bins it before it touches the next part.
- **Bare anvil:** nothing on the anvil plate and nothing under the jaws.
- **Spill:** a part outside the rack, the anvil and both bins. It is left where it falls and cleared in the reset, never on camera.
- **Quarter turn:** the most any object is tilted in the gripper. Standing a part up and laying it on its side both stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Check the caliper at zero

Goal: the jaws shut on nothing, a settled reading of zero in frame, and the jaw parked again.

The zero check happens once, before the first part. If the caliper is not at zero, every reading in the episode is wrong, so this is checked before anything is measured.

1.1 Close the jaws

- **Check:** the anvil is bare, with nothing standing on it and nothing under the jaws.
- With the **gripper**, close on the slider knob square, and lower it slowly until the two jaws meet and the slider stops.
- Open the gripper and take it right away from the caliper and the stand.

Check: the jaws are shut with no daylight between them and nothing is trapped between them.

1.2 Read the zero

- Hold still and wait until the number on the display stops changing for 2 seconds.
- The display has to read zero.

Check: the display reads zero and is readable in frame. If it reads anything else, end the episode and zero the caliper by hand during reset. Do not press a button to zero it on camera.

1.3 Park the jaw

- With the **gripper**, close on the slider knob square and raise the slider until the moving jaw stands clear above the height of the tallest part.
- Open the gripper and lift clear.

Check: the jaw is parked high, the anvil is bare, and the slider does not creep back down. One zero check, and one only. It is not run again between parts.

Step 2: Stand the part on the anvil

Goal: one part standing upright in the middle of the anvil, square and steady, and its rack slot empty.

Parts are worked in slot order, slot 1 first and slot 5 last. The rack and the stand both grip the table, so nothing holds either of them while the part is moved.

2.1 Take the part from its slot

- With the **gripper**, close on the part square on its long sides, well below its top end, and lift it straight up out of the slot until it is clear of the rack.
- Keep it upright. Do not lean it over on the way out.

Check: the part hangs upright in the gripper and does not slip. If it hangs tilted, stand it back in its slot and take it again. Make no more than two corrective takes. If it still will not hang upright, end the episode and correct the part during reset.

2.2 Stand it on the anvil

- Carry the part forward to the anvil, keeping it upright and clear of the caliper, and bring it in under the parked jaw.
- Lower it until it takes its own weight on the middle of the anvil, then open the gripper and lift clear.

- Only one part is on the anvil at a time.

Check: the part stands upright in the middle of the anvil, flat on its end, square to the jaw, touching nothing but the anvil, and it does not rock or lean. If it rocks or leans, lift it clear and stand it again, once.

Step 3: Measure the tall way and mark it

Goal: the jaw resting on the top of the standing part, a settled reading in frame, and one X in the matching box on the TALL strip of the part's row.

The jaw stays down on the part while the mark is made, so the display keeps showing the reading that is being written down.

3.1 Bring the jaw down

- With the **gripper**, close on the slider knob square, and lower it slowly until the moving jaw meets the top of the part and the arm feels it stop.
- Stop there. Do not push the jaw down past the part.
- Open the gripper and take it right away from the caliper, the stand and the part.

Check: the jaw rests flat on the top of the part, the part still stands square and has not been pushed over or slid across the anvil, and nothing is touching the caliper.

3.2 Read the display

- Hold still and wait until the number on the display stops changing for 2 seconds.
- That number is the tall way reading.
- The camera records the reading. It is not written down anywhere but the log sheet.

Check: the display is readable in frame and the number has held still. If the number will not settle, end the episode and let review flag it.

3.3 Mark the TALL strip

- With the **gripper**, close on the marker barrel, lift it straight up out of its holder, and bring the tip over the log sheet.
- Go to the row with the part's slot number on it. Look along the TALL strip and find the one box whose printed range holds the number on the display.
- Draw one X in that box, fully inside its lines.
- With the **gripper**, carry the marker back to its holder, lower it in until it stands on its own, release, and lift clear.

One X, in one box, on the right row. The slider is not touched while the marker is out.

Check: there is one X on the TALL strip of the right row, it is inside its box, and no other box on that strip is marked. A wrong mark is not rubbed out and not drawn over. Leave it and let review flag the episode.

Step 4: Measure the wide way and mark it

Goal: the part lying on its side under the jaw, a settled reading in frame, one X in the matching box on the WIDE strip, and one X in PASS or FAIL on the same row.

The part is turned a quarter turn and no more. It is never turned over in the air and it is never rolled across the anvil.

4.1 Park the jaw and lay the part down

- With the **gripper**, close on the slider knob square, raise the slider until the jaw is parked clear above the part, then open the gripper and lift clear.
- With the **gripper**, close on the part square, lift it straight up off the anvil, turn it a quarter turn so it is on its side, and lower it back onto the middle of the anvil.
- Open the gripper and lift clear.

Check: the part lies flat on its side in the middle of the anvil, square to the jaw, touching nothing but the anvil, and it does not rock. If it rocks or sits off the middle, lift it clear and lay it down again, once.

4.2 Bring the jaw down and read

- With the **gripper**, close on the slider knob square, and lower it slowly until the moving jaw meets the top of the part and the arm feels it stop.
- Stop there. Do not push the jaw down past the part.
- Open the gripper, take it right away, hold still, and wait for the number to settle.
- That number is the wide way reading.

Check: the jaw rests flat on the part, the part has not been squeezed out of place, and the display is readable in frame.

4.3 Mark the WIDE strip

- With the **gripper**, take the marker out of its holder and bring the tip over the log sheet.
- On the same row, look along the WIDE strip and find the one box whose printed range holds the number on the display.
- Draw one X in that box, fully inside its lines.

Check: there is one X on the WIDE strip of the right row and no other box on that strip is marked.

4.4 Mark pass or fail

- Look at the two X marks on the row, the one on the TALL strip and the one on the WIDE strip.
- **If both X marks are in heavy boxes**, draw one X in the PASS box on that row.
- **If either X mark is in a plain box**, draw one X in the FAIL box on that row.

- One result box is marked on a row, never both.
- With the **gripper**, carry the marker back to its holder, lower it in until it stands on its own, release, and lift clear.

The result is read off the sheet, not from memory. Two heavy boxes is a pass. Anything else is a fail.

Check: the row holds three X marks, one on each strip and one result, each fully inside its own box, and the marker stands in its holder. A wrong result is not rubbed out and not drawn over.

Step 5: Put the part in its bin

Goal: the part standing in the bin its row says, and the anvil bare again.

Both bins stand in their holders, so a part can be dropped into either one without anything holding it. Neither bin is ever lifted.

5.1 Park the jaw

- With the **gripper**, close on the slider knob square, raise the slider until the jaw is parked clear, then open the gripper and lift clear.

Check: the jaw is parked high enough for the part to come out from under it freely.

5.2 Lift the part off the anvil

- With the **gripper**, close on the part square, and lift it straight up off the anvil until it is clear of the jaw.

Check: the part is clear of the caliper and the gripper has not knocked the stand.

5.3 Drop it in its bin

- **If the row is marked PASS**, carry the part off the anvil **to the right** to the pass bin.
- **If the row is marked FAIL**, carry the part off the anvil **to the left** to the fail bin.
- Bring it over the bin mouth, low to the rim, stop, and open the gripper so it drops straight in.
- A part never passes over the bin it is not going into.

Check: the anvil is bare and the part is in the bin its row says. If a part has gone into the wrong bin, leave it there and let review flag the episode. Parts are never taken back out of a bin.

Step 6: Do the next part

- Come back empty to the rack and take the part from the next slot in order.
- Run Steps 2 to 5 again for that part, marking its readings on the row with its slot number.
- Five parts, worked one at a time from slot 1 to slot 5. A part is finished and in a bin before the next part is touched.
- The zero check is not run again. The caliper is checked once per episode.

Check: after each part, the rack has one fewer part in it, the anvil is bare, one more row is fully marked, and no row is skipped or marked twice.

Step 7: End the episode

- Confirm the rack: all five slots empty.
- Confirm the anvil: bare, with the jaw parked clear above it.
- Confirm the bins: three parts in the pass bin, two parts in the fail bin, both bins upright in their holders.
- Confirm the log sheet: five rows, three X marks on each row, each mark inside its own box, and three rows marked PASS and two rows marked FAIL.
- Confirm the tools: the marker upright in its holder, the good size card unmoved.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the parts out of both bins and stand the bins back in their holders, empty and dry.
2. Count the parts back into the rack: exactly five, three good, one over size the tall way and one under size the wide way. Replace any part that has been chipped, dented or burred so the five are the same five next time.
3. Shuffle the five parts into the five slots so they do not sit in the same order twice in a row, then stand each one upright, tall way up.
4. Pick up any part that fell off the anvil or off the table and put it back in the rack. Replace it if it is damaged.
5. Wipe the log sheet clean and dry it, so all five rows start blank.
6. Wipe any dust off the anvil, the caliper jaws and the tabletop, and dry them.
7. Close the caliper jaws by hand and check the display reads zero. Zero it by hand if it does not, and replace the battery if the display is faint.
8. Raise the slider and leave the jaw parked clear above the height of the tallest part.
9. Stand the marker in its holder with the cap off. Replace a marker that is running dry.
10. Confirm the rack, the caliper stand, both bins and the log sheet all still grip the table, and that the caliper is still clamped tight in its stand.
11. Confirm the good size card still lies flat beside the stand and is still readable.
12. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a part is measured before the zero check, the wide way is measured before the tall way is marked, a part goes in a bin before both of its readings are marked, the result box is marked before both strip marks are on the row, a new part is taken while the last one is still on the anvil, or a step is skipped.
- **SOP rule broken:** Steps 1 to 6, check zero once, then for each part stand it, measure the tall way, mark it, measure the wide way, mark it, mark the result, bin it, in that order.
- **Coaching note:** zero first, then one part at a time, all the way through. Mark the number while it is still on the display.

Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the rack, the stand, a bin or the log sheet, or moves in front of the caliper display or the log sheet. Any object taken or carried by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 7, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.
- **Coaching note:** one arm only. If a second arm has worked, the episode is a bimanual episode wearing the wrong label.

Violation: Object held or steadied

- **Visible cue:** the gripper presses on the rack, the caliper stand, a bin or the log sheet to keep it still while working, or holds the part on the anvil while the reading is taken, instead of letting the holder and the anvil do it.
- **SOP rule broken:** Steps 1 to 5, every object grips the table or stands in its holder, and nothing on this station is held.

- **Coaching note:** if something needs holding, the holder is wrong. Fix the station, do not add a hand.

Violation: Zero check skipped, or run more than once

- **Visible cue:** the first part goes on the anvil with no zero check on camera, the jaws are closed on something during the zero check, the zero reading is called before the number holds still, or the jaws are closed on nothing again between parts.
- **SOP rule broken:** Step 1, one zero check with a bare anvil, before the first part, and one only.
- **Coaching note:** check it once, at the start, on nothing. A mid episode zero check means the readings before it are being doubted.

Violation: Caliper button pressed, or the caliper moved in its stand

- **Visible cue:** the gripper presses a button on the caliper, or the caliper turns, slips or lifts in its clamp while the slider is worked.
- **SOP rule broken:** Materials checklist and Steps 1 to 5, the caliper is zeroed by hand during setup and the gripper touches nothing on it but the slider knob.
- **Coaching note:** the knob is the only handle. A caliper that moves in the stand makes every reading after it meaningless.

Violation: Jaw forced down on the part

- **Visible cue:** the slider keeps going down after the jaw meets the part, the part is squeezed, pushed over, slid across the anvil or lifted off it by the jaw, or the slider is dropped onto the part instead of lowered.
- **SOP rule broken:** Steps 3.1 and 4.2, lower the slider slowly and stop the moment the jaw meets the top of the part.
- **Coaching note:** let it touch, then stop. A forced jaw reads small and moves the part.

Violation: Gripper still on the caliper while the number settles

- **Visible cue:** the gripper stays closed on the slider knob, rests on the caliper, the stand or the part, or nudges any of them while the number is settling.
- **SOP rule broken:** Steps 1.2, 3.2 and 4.2, open the gripper and take it right away before waiting for the reading.
- **Coaching note:** let go, back off, then read. A hand on the caliper is a hand on the number.

Violation: Reading called before it settles

- **Visible cue:** a mark is made while the number on the display is still changing, or before it has held still for 2 seconds.
- **SOP rule broken:** Steps 3.2 and 4.2, the reading is the number that has stopped changing and held still for 2 seconds.

- **Coaching note:** wait for it to stop. A moving number is not a measurement.

Violation: A dimension skipped, or measured twice

- **Visible cue:** a part is binned after only one measurement, the part is never turned onto its side, the same dimension is measured a second time to check it, or a part goes back on the anvil after it has been binned.
- **SOP rule broken:** Steps 3 and 4, two dimensions per part, the tall way then the wide way, each measured once.
- **Coaching note:** two numbers per part, one look each. If a reading looked wrong, tag it and let review decide.

Violation: Part not standing or lying square on the anvil

- **Visible cue:** the part leans, rocks, sits off the middle of the anvil, hangs over its edge, or leans on the caliper beam when the jaw comes down.
- **SOP rule broken:** Steps 2.2 and 4.1, the part sits square in the middle of the anvil, touching nothing but the anvil.
- **Coaching note:** square it before you measure it. A leaning part gives a long number.

Violation: Part rolled, pushed or turned over

- **Visible cue:** the part is rolled, dragged or pushed across the anvil to turn it, tipped over by the jaw, or rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Step 4.1, the part is lifted clear, turned a quarter turn in the gripper, and set back down.
- **Coaching note:** lift it to turn it. A pushed part tells the model nothing about grasping.

Violation: More than one part handled at a time

- **Visible cue:** the gripper closes on two parts at once, a second part is taken out of the rack before the first is binned, or two parts sit on the anvil together.
- **SOP rule broken:** Steps 2 to 6, one part at a time, taken, measured, marked and binned before the next one is touched.
- **Coaching note:** one part, one trip through the steps. A second part on the table loses the row it belongs to.

Violation: Parts worked out of slot order

- **Visible cue:** a part is taken from a slot out of turn, or the row marked is not the row with that part's slot number on it.
- **SOP rule broken:** Steps 2.1 and 6, parts are worked slot 1 through slot 5 in order, and each part is marked on the row with its slot number.
- **Coaching note:** left to right, top to bottom. The row number is the slot number.

Violation: Wrong box marked

- **Visible cue:** the X goes in a box whose printed range does not hold the number on the display, the X goes on the wrong strip, the wrong row is marked, two boxes on one strip are marked, or a strip is left blank.
- **SOP rule broken:** Steps 3.3 and 4.3, one X on each strip, in the one box whose range holds the reading, on the row with the part's slot number.
- **Coaching note:** read the number, then find the box that holds it. One reading matches one box.

Violation: Pass or fail marked wrong

- **Visible cue:** PASS is marked on a row that has an X in a plain box, FAIL is marked on a row whose two X marks are both in heavy boxes, both result boxes are marked, or the result box is left blank.
- **SOP rule broken:** Step 4.4, both X marks in heavy boxes is a PASS, and anything else is a FAIL.
- **Coaching note:** look at the row, not at the part. Two heavy boxes pass, anything else fails.

Violation: Mark rubbed out, drawn over, or outside its box

- **Visible cue:** a mark is wiped off, drawn over, corrected, or a mark crosses or sits outside a box line so it cannot be read.
- **SOP rule broken:** Steps 3.3, 4.3 and 4.4, each mark is one X fully inside its own box, and a wrong mark is tagged, never rubbed out.
- **Coaching note:** leave the mistake and let review flag it. A corrected sheet cannot be scored.

Violation: Part put in the wrong bin

- **Visible cue:** a part marked PASS goes in the fail bin, a part marked FAIL goes in the pass bin, or a part is left on the anvil or on the table instead of a bin.
- **SOP rule broken:** Step 5.3, the part goes in the bin its own row says.
- **Coaching note:** read the row, then move. The sheet decides the bin, not the eye.

Violation: Part taken back out of a bin

- **Visible cue:** a part is lifted back out of a bin, or a bin is tipped, to move a part that was put in the wrong place.
- **SOP rule broken:** Step 5.3, parts are never taken back out of a bin, and a wrong bin is tagged, not fixed.
- **Coaching note:** once it is in, it stays in. A correction pick makes the episode unlearnable.

Violation: Part released before it is over its bin

- **Visible cue:** the gripper opens while the part is still over the anvil, over open tabletop, or over the wrong bin, or it opens high so the part bounces out.
- **SOP rule broken:** Step 5.3, the gripper opens low over the mouth of the bin the part belongs in.
- **Coaching note:** carry it all the way, low to the rim, then open. A part dropped early is a part on the floor.

Violation: A part carried over the wrong bin

- **Visible cue:** a pass part passes over the fail bin, or a fail part passes over the pass bin, on its way to its own bin.
- **SOP rule broken:** Step 5.3, a pass part leaves the anvil to the right and a fail part leaves the anvil to the left.
- **Coaching note:** route around, not across. A part dropped over the wrong bin lands in the wrong bin.

Violation: Good size card touched or moved

- **Visible cue:** the gripper picks up, slides or knocks the good size card, or lays a part on it to compare.
- **SOP rule broken:** Materials checklist and Step 4.4, the good size card is looked at and never touched.
- **Coaching note:** compare with the eyes. The card is a reference, not a work surface.

Violation: Tool not returned to its home

- **Visible cue:** the marker is left lying on the log sheet, on the anvil, in a bin or on the table, the gripper opens while the marker is still above its holder, or the marker is still in the gripper when the next step begins.
- **SOP rule broken:** Steps 3.3 and 4.4, the marker returns to its holder and stands in it before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower until it touches, then release.

Violation: Spill cleaned up

- **Visible cue:** after a part lands on the table or on the floor, the gripper is used to sweep, push, or pick it up.
- **SOP rule broken:** Steps 1 to 6, a spill is left where it falls and is cleared in the reset.
- **Coaching note:** cleanup motions are not part of the task. The spill is data, the cleanup is noise.

Violation: Bad pick, carry, or release

- **Visible cue:** an object is dragged along the table with no daylight under it, the gripper shifts or seats its hold again while the object is off the table, a part hangs tilted in the gripper because the grasp was taken off square, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 5, one object at a time, lifted straight up clear of the surface before any horizontal motion, gripped square, and tilted no more than a quarter turn.
- **Coaching note:** set it down to change the grip, and lift before the carry. A part that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a part, a bin, the marker, the rack or the log sheet is shoved, nudged, or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 6, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, the gripper or a carried object strikes the caliper, the stand, the anvil, the rack, a bin or the log sheet and knocks, tips, slides, or displaces it; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 6, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. The caliper stands in the middle of everything, so the route matters more than the reach.

Violation: Manipulation outside the camera frame

- **Visible cue:** a part pick, a placement on the anvil, a slider move, a reading or a mark happens partly or wholly outside the environment camera frame, or the arm blocks the camera's view of the caliper display or the log sheet while the reading is being taken or marked.
- **SOP rule broken:** Steps 1 to 6, every manipulation happens inside the camera frame and the display and the sheet stay visible.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and keep the arm clear of the display.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a part still in the rack or on the anvil, the jaw still down, a row unmarked or part marked, a part in the wrong bin, the marker out of its holder, the arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 7, confirm the rack, the anvil, the bins, the log sheet and the tools, then return the arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective caliper: one that will not hold zero, whose number never settles, whose display is faint or unreadable in frame, or whose slider sticks, creeps or drops under its own weight.
- Defective caliper stand: one that slides or lifts on the table under a correct slider move, or whose clamp lets the caliper turn or slip.
- Defective slider knob: one the gripper cannot close on square, or one that turns on the slider instead of moving it.
- Defective anvil: one that is not level, that rocks, or that is too small for a part to lie on its side on it.
- Defective rack: one that slides, turns or lifts under a correct pick, or a slot that grips a part so a straight lift will not free it.
- Defective bin: one that will not stand upright in its holder, or that is dented or cracked.
- Dry marker, or a log sheet that will not wipe clean so a box starts the episode already marked.
- A log sheet printed wrong: overlapping ranges, a missing heavy outline, boxes that touch, or ranges that do not match the good size card.
- A part so close to the edge of a printed range that the reading lands on a box line and no single box holds it. Replace the part before the next episode.
- Rack holding fewer or more than five parts at episode start, or the wrong mix of them. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Close the caliper jaws for the zero check
2. Raise the slider to park the jaw
3. Take one part from its rack slot
4. Stand one part upright on the anvil
5. Lower the slider onto the part
6. Take the marker from its holder
7. Draw one X on the TALL strip
8. Lift and turn the part onto its side
9. Draw one X on the WIDE strip
10. Draw one X in the PASS or FAIL box
11. Return the marker to its holder
12. Lift one part off the anvil
13. Drop one part in the pass bin
14. Drop one part in the fail bin
15. Return the arm home and end the episode